BLOCKCHAIN

P2P Networks Visual Guide

DHTs, BitTorrent swarming, gossip,
and NAT traversal — systems without servers.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Why Peer-to-Peer?

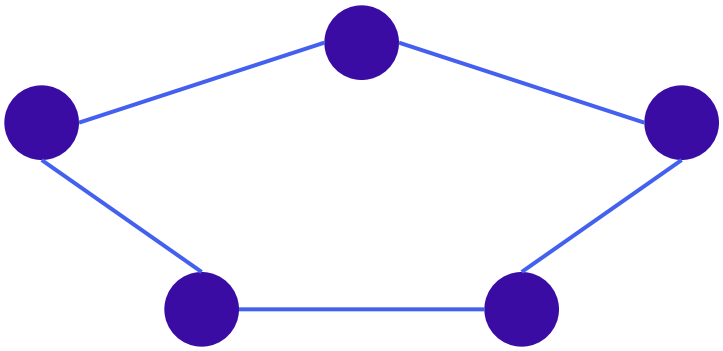
No center to fail, censor, or pay for — capacity grows with users.

CLIENT-SERVER

server = bottleneck + single point of failure

PEER-TO-PEER

every node is client AND server



Tradeoff: you gain resilience and scale, you lose easy coordination and consistent views.

STEP 2/6

Peer Discovery & Bootstrapping

The chicken-and-egg problem: finding your first peer.

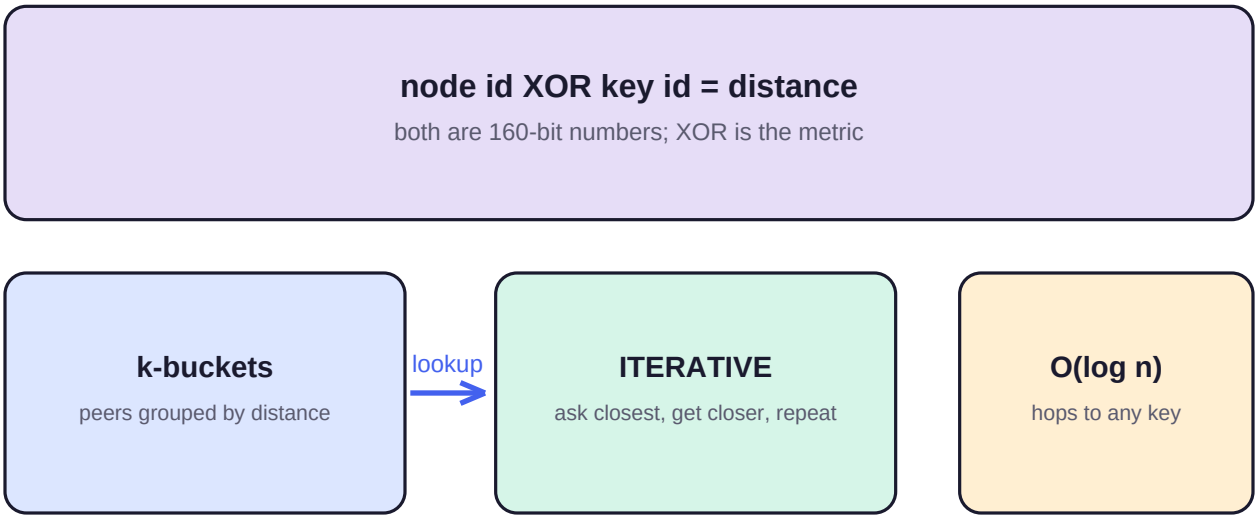
Bootstrap nodes	hardcoded well-known peers — the on-ramp
Peer exchange	ask peers for their peer lists — the network teaches itself
mDNS / LAN	local discovery without any server at all
DHT bootstrap	once in the DHT, find everything by key

Bootstrap nodes are the last centralization — eclipse attacks target this step.

STEP 3/6

Kademlia DHT

A distributed hash table: keys map to the 'closest' nodes by XOR distance.



BitTorrent, IPFS, and Ethereum discovery all run Kademlia variants. It just works at planetary scale.

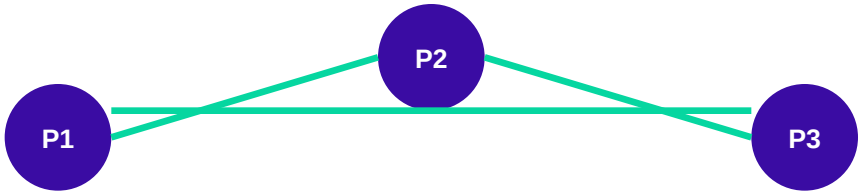
STEP 4/6

BitTorrent Swarming

Split the file; everyone uploads what they have. Downloaders become uploaders.

FILE -> PIECES

e.g. 1 GB -> 4096 x 256 KB pieces, each hash-verified



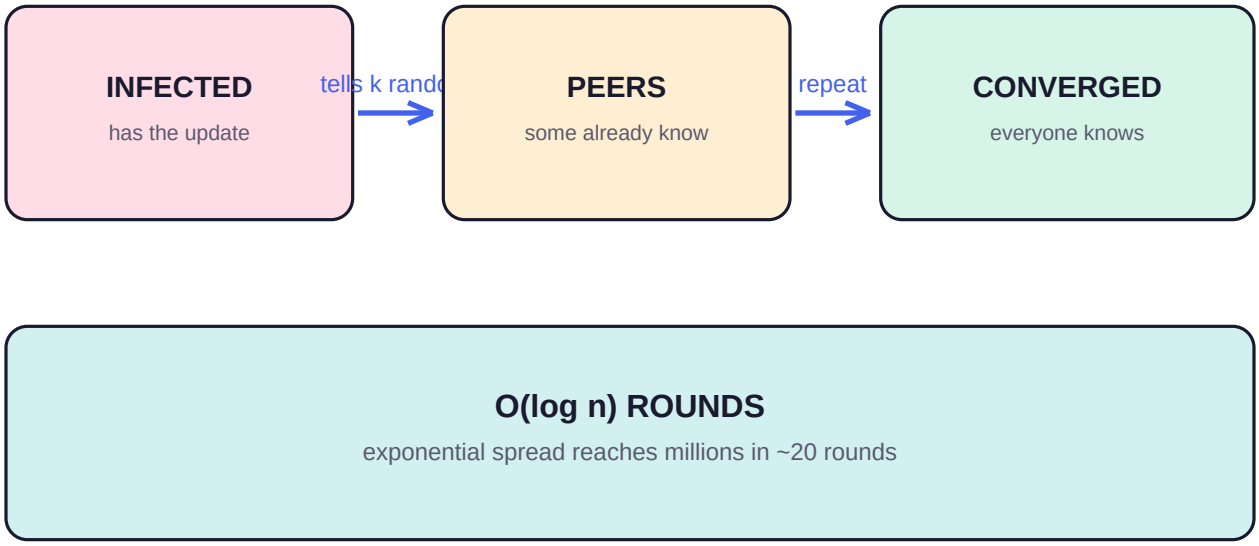
Tit-for-tat: upload to those who upload to you. Rarest-first piece selection.

Aggregate bandwidth grows with the swarm — the opposite of a server melting under load.

STEP 5/6

Gossip Protocols

Rumors spread exponentially: each node tells a few random peers, repeatedly.



Eventual consistency with zero coordination — blockchains and membership protocols gossip constantly.

STEP 6/6

NAT Traversal

Most peers hide behind NATs. Getting two of them to connect directly takes tricks.

Hole punching	both dial out simultaneously; NATs open return paths (STUN)
Relay (TURN)	when punching fails, bounce through a relay — slower but reliable
UPnP / PCP	ask the router to forward a port — works on home networks
ICE	try everything in order: host -> reflexive -> relayed

STEP ✓/✓

Key Takeaways

P2P Networks — Remember

- P2P trades easy coordination for resilience: capacity grows with the user base.
- Bootstrapping solves peer discovery; guard it against eclipse attacks.
- Kademlia DHTs route by XOR distance in O(log n) hops — the backbone of BitTorrent/IPFS.
- BitTorrent swarms split files into verified pieces; tit-for-tat aligns incentives.
- Gossip protocols spread updates exponentially with zero coordination.
- NAT traversal stacks hole punching, relays, and ICE to connect peers directly.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev